Candidate Guidelines

Setup checklist for the Data Science Hackathon

Please set up everything on your own laptop before the hackathon. The hackathon is time-bound, so make sure everything works in your environment beforehand rather than spending time on installation or setup issues on the day.

1. Laptop requirements

- Windows or Mac laptop with at least 8 GB RAM (16 GB recommended) and 5 GB of free disk space.
- Please bring your laptop charger and make sure you have a stable internet connection.

2. Install Python and Jupyter Notebook

- Install Anaconda (https://www.anaconda.com/download), which includes Python and Jupyter Notebook. Alternatively, Python 3.10 or above with Jupyter Notebook / VS Code is fine.
- Open Jupyter Notebook once and confirm that a new notebook opens and runs.

3. Install the required Python libraries

Run the following command in Anaconda Prompt (Windows) or Terminal (Mac):

```
pip install pandas numpy scikit-learn lightgbm xgboost imbalanced-learn
    matplotlib seaborn plotly pyarrow joblib
```

Then run this in a notebook cell to confirm everything works:

```
import pandas, numpy, sklearn, lightgbm, xgboost, imblearn, matplotlib, seaborn, plotly, joblib
print("All libraries installed successfully")
```

4. Install a dashboard tool: Power BI (Windows) or Tableau (Mac)

- Windows users: download Power BI Desktop for free from the Microsoft Store or https://powerbi.microsoft.com/desktop.
- Mac users: Power BI Desktop does not run on Mac, so please install Tableau instead, either Tableau Public (free, https://public.tableau.com) or Tableau Desktop (free for students through the Tableau for Students programme, or a free trial).
- Open the tool once and load any small CSV file to confirm it works.
- If you are new to the tool, please go through the basics beforehand: loading data, creating relationships between tables, creating measures (DAX in Power BI, calculated fields in Tableau), and building charts, filters and multi-page reports / dashboards. Free starter tutorials:
  - Power BI: https://learn.microsoft.com/power-bi/fundamentals/desktop-getting-started
  - Tableau: https://www.tableau.com/learn/training

5. GitHub account

- Please create a GitHub account (if you don't have one) so you can submit your work as a repository link. A zipped folder is also accepted.

6. Use of AI tools

- Using AI tools is encouraged for building the model and code, and for developing the Power BI / Tableau dashboard. Please set up and log in to the AI tools you plan to use before the hackathon.

7. What we are evaluating

- The aim is to see how well you understand the end-to-end data science process and how fully you cover the expectations of the use case. Visual polish and depth in any one area matter less.
- Please read the attached use case document fully before the hackathon so you understand the expected deliverables, folder structure and submission format.